Vision of the Department

Tobeawell-knowncentreforpursuingcomputereducationthroughinnovativepedagogy,value-basededucationandindustrycollaboration.

Mission of the Department

ToestablishlearningambienceforusheringincomputerengineeringprofessionalsincoreandmultidisciplinaryareabydevelopingProblem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dreamofwhereyouwant.	**Mission:** Means to achieveVision

ProgramEducationalObjectivesoftheprogram(PEO):(broadstatementsthatdescribethe professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CLabbreviation pronounceasPep-si-lL easy to recall**
PEO2	**CoreCompetence**	**E: Environment (LearningEnvironment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C:Core Competence**	
PEO5	**Learning Environment**	**L:Breadth(Learning in diverse areas)**	

ProgramOutcomes(PO):(statementsthatdescribewhatastudentshouldbeabletodoandknowby the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"Iamanengineer,andIknowhowtoapplyengineeringknowledgetoinvestigate,analyseanddesign solutions to complex problems using tools for the entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity:I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Srushti Chaudhari

NameandSignature ofStudentand Date
(Signature and Date in Handwritten)

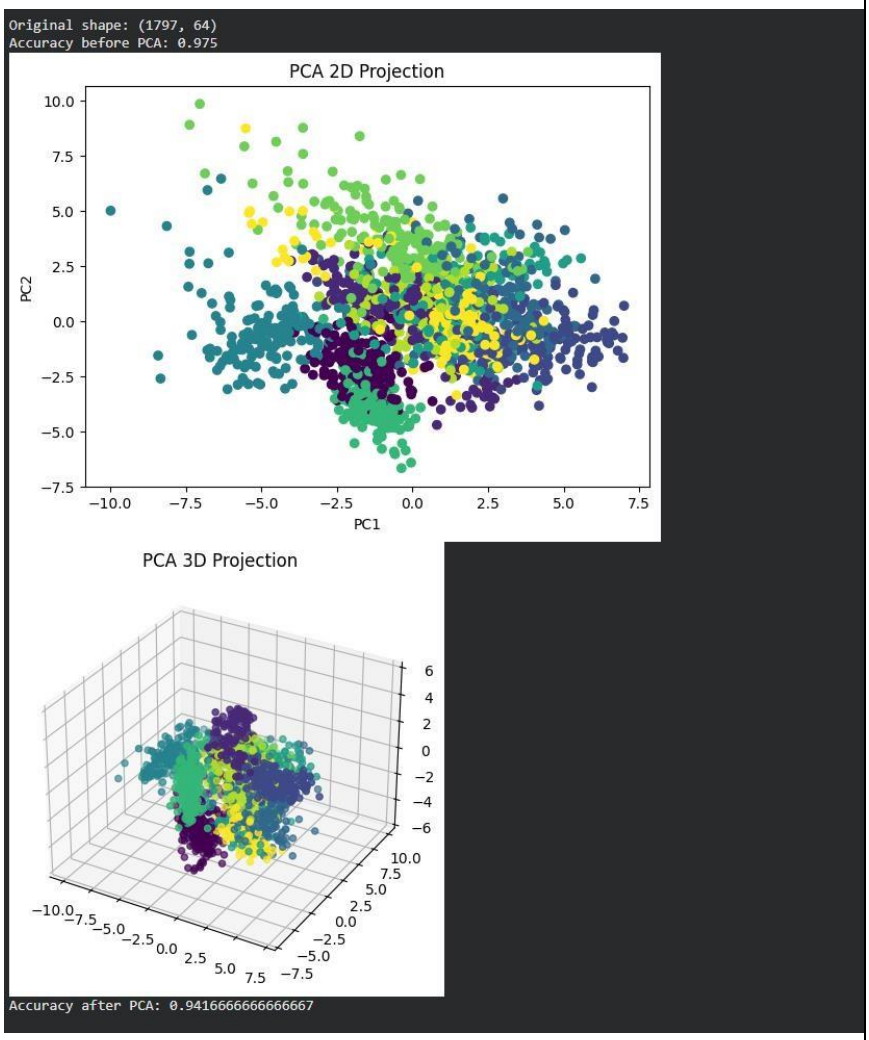
Session	**2025-26 (Even)**	**Course Name**	**Lab:AIML**
Semester	**6**	**Course Code**	**23CT1602**
Roll No	**23**	**Name of Student**	**Srushti Chaudhari**

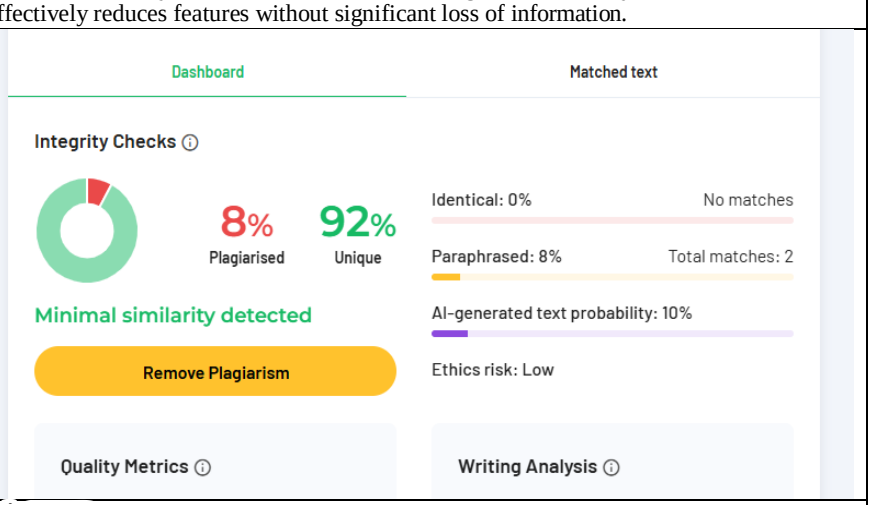
Practical Number	**8**
Course Outcome	CO-1
Aim	To reduce dimensionality using Principal Component Analysis (PCA).
Problem Definition	To apply Principal Component Analysis (PCA) on a high-dimensional dataset to reduce the number of features while preserving maximum variance, visualize the 2D/3D projections, and compare model performance before and after PCA.
Theory (100 words)	Principal Component Analysis (PCA) is a dimensionality reduction technique used to reduce the number of features in a dataset while retaining most of the important information. PCA transforms the original features into a new set of features called Principal Components: • These components are uncorrelated • Ordered by maximum variance • First component contains highest variance • Second component contains second highest variance Steps of PCA 1. Standardize the dataset 2. Compute covariance matrix 3. Calculate eigenvalues and eigenvectors 4. Sort eigenvalues in descending order 5. Select top K principal components 6. Transform data into new feature space Advantages • Reduces dimensionality • Removes multicollinearity • Improves model performance • Reduces overfitting • Faster training Applications • Image compression • Face recognition • Data visualization • Noise reduction • Feature extraction

Procedure and Execution (100Words)	PS 1 :- Import required libraries Load high-dimensional dataset Standardize the dataset Apply PCA to reduce dimensions Transform data into 2D and 3D Visualize PCA projections Train model before PCA Train model after PCA Compare performance Analyze results

```
# Practical 8: PCA

# --------------------------------
# Import libraries
# --------------------------------
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_digits
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

from mpl_toolkits.mplot3d import Axes3D

# --------------------------------
# Load high dimensional dataset
# --------------------------------
digits = load_digits()
X = digits.data
y = digits.target

print("Original shape:", X.shape)

# --------------------------------
# Train test split
# --------------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# --------------------------------
# Model BEFORE PCA
# --------------------------------
model = LogisticRegression(max_iter=5000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
before_acc = accuracy_score(y_test, y_pred)

print("Accuracy before PCA:", before_acc)

# --------------------------------
# Standardize data
# --------------------------------
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# --------------------------------
# Apply PCA (2D)
# --------------------------------
pca_2 = PCA(n_components=2)
```

```
X_pca_2 = pca_2.fit_transform(X_scaled)

# --------------------------------
# 2D Visualization
# --------------------------------
plt.figure(figsize=(7,5))
plt.scatter(X_pca_2[:,0], X_pca_2[:,1], c=y, cmap='viridis')
plt.title("PCA 2D Projection")
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.show()

# --------------------------------
# Apply PCA (3D)
# --------------------------------
pca_3 = PCA(n_components=3)
X_pca_3 = pca_3.fit_transform(X_scaled)

# --------------------------------
# 3D Visualization
# --------------------------------
fig = plt.figure(figsize=(7,5))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(X_pca_3[:,0], X_pca_3[:,1], X_pca_3[:,2], c=y)
ax.set_title("PCA 3D Projection")
plt.show()

# --------------------------------
# Apply PCA for training
# --------------------------------
pca = PCA(n_components=20)
X_pca = pca.fit_transform(X_scaled)

X_train_pca, X_test_pca, y_train, y_test = train_test_split(
    X_pca, y, test_size=0.2, random_state=42
)

# --------------------------------
# Model AFTER PCA
# --------------------------------
model2 = LogisticRegression(max_iter=5000)
model2.fit(X_train_pca, y_train)

y_pred2 = model2.predict(X_test_pca)
after_acc = accuracy_score(y_test, y_pred2)

print("Accuracy after PCA:", after_acc)
```

	Output: 
OutputAnalysis	The PCA algorithm reduced the high-dimensional dataset into lower dimensions. The 2D and 3D visualizations showed that the data points were clearly separated. The model accuracy before PCA was calculated using original features. After applying PCA, the dimensionality was reduced significantly. The accuracy after PCA remained similar or slightly improved. This shows PCA preserved important information while reducing features.
Link of student Githubprofilewhere lab assignment has	https://github.com/SRchaudhari15/AIML-Practicals.git

been uploaded	
Conclusion	Principal Component Analysis (PCA) was successfully implemented to reduce dimensionality of high-dimensional data. The dataset was transformed into 2D and 3D projections for visualization. The performance comparison before and after PCA showed that dimensionality was reduced while maintaining model accuracy. Hence, PCA effectively reduces features without significant loss of information.
Plag Report (Similarityindex<12 %)	
Date	9/3/26

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